

SIASET - MVP

“The eye that understands life”

1) PCR

Introduction

DNA exists in extremely small amounts — too little to study, detect, or measure directly.

PCR is a technique that **selects one specific DNA region** and **copies it millions of times**.

By cycling temperature, PCR repeatedly separates DNA strands, allows primers to bind, and extends new DNA copies.

This process is powerful but highly sensitive — **small changes in conditions can completely change the result**.

In this lab, you will control the reaction conditions and observe how your decisions affect the final DNA product.

Research

A) Scope

Why is PCR used ?

Biology works at a scale we cannot see or measure directly.

- DNA exists in **extremely tiny amounts**
- Most techniques (sequencing, cloning, diagnosis) **require a lot of DNA**
- Cells don't give you enough of a specific sequence on demand

so how do we make millions of copies of one specific DNA region ?

PCR is the answer

- Before PCR , DNA cloning required bacteria , plasmids and days to weeks worth of time. The diagnostics weren't any better as they were slow , inaccurate and often impossible to achieve due to low DNA amounts.

- What PCR actually does is that it selects a specific region in the DNA using primers , it copies that region then amplifies it exponentially ; giving millions of copies of that specific region.

- **(1) Detection: How does PCR change biology and diagnostics ?**

PCR answers yes/no questions with extreme accuracy and sensitivity , Example : (Is there a virus present ? - Is this gene in the genome ? - Is this mutation present ?).

PCR is used in:

- COVID tests
- HIV detection
- TB diagnosis

- **(2) Identification: Whose DNA is that ?**

PCR allows you to amplify unique regions.

Used in:

- Forensics
- Paternity testing
- Ancestry testing

- Without PCR the DNA quantity would be too low to identify such things.

- **(3) Preparation: I need DNA to do something else.**

Most molecular biology techniques depend on PCR first. PCR is not always the final step , it's often step zero.

- DNA sequencing

- Cloning a gene
- CRISPR guide validation
- Mutagenesis

They all start with PCR as the first step.

- **(4) Quantity: How much is there ?**

With **qPCR / RT-qPCR**

- You don't just amplify
- You **measure** DNA or RNA levels

Used in:

- Gene expression studies
- Cancer research
- Drug testing

Where students fail most ?

These are the moments where:

- **"I followed the steps but it didn't work"**
- **"I don't understand why there's no band"**
- **"The theory makes sense but the result doesn't"**

This is where our virtual lab shines most.

- **(1) Primer design (#1 Failure)**

What students THINK:

- "Primers just need to match the gene."

What actually happens:

- Small design flaws completely kill PCR.

Common student failures:

- Primers too short / too long
- GC content too high or too low
- 3' end mismatches
- Primers binding multiple sites
- Primer-dimers

Why students fail here:

- They don't visualize binding
- They underestimate specificity

How this appears in real life:

- No band
- Multiple bands
- Strong primer-dimer band

- **(2) Annealing temperature misunderstanding**

What students THINK:

- "Lower temperature = better binding."

Reality:

- Too low = nonspecific binding
- Too high = no binding

Typical student mistake:

- Randomly choosing temperatures
- Not relating it to primer T_m

Real lab result:

- Smears
- Multiple bands
- No product

Why this is hard:

- Temperature feels abstract

- The effect is invisible until the gel

- **(3) Interpreting gels incorrectly**

What students THINK:

- “A band means success.”

Reality:

- Not all bands are good
- Size matters
- Intensity matters

Common failures:

- Confusing primer-dimers with real product
- Ignoring ladders
- Misreading smears

Educational gap:

- Students rarely see bad gels explained

- **(4) Assuming PCR is perfectly exponential**

What students THINK:

- “Each cycle doubles DNA forever.”

Reality:

- Efficiency drops
- Plateau phase exists

Consequences:

- Endpoint PCR is misleading
- Band brightness $\neq$ starting quantity

This misunderstanding blocks:

- qPCR understanding

- Experimental design

- **(5) Not understanding controls**

Common student mindset:

- “If there’s no band, PCR failed.”

Reality:

- Maybe template was missing
- Maybe primers are wrong
- Maybe contamination exists

Students fail to:

- Use positive controls
- Use negative controls
- Interpret control outcomes

This is huge in diagnostics.

- **(6) Treating PCR like a recipe**

What students do:

- Memorize steps
- Don’t reason

Why this fails:

- PCR requires decision-making
- Every change affects outcome

Our simulator should punish blind following and reward understanding.

What parameters affect outcomes ?

The main parameters that affect PCR outcomes are :

1. **Template DNA (starting material)**
 2. **Primer design (MOST IMPORTANT PARAMETER)**
 3. **Annealing temperature**
 4. **Mg²⁺ concentration**
 5. **DNA polymerase type**
 6. **Extension time**
 7. **Number of cycles**
 8. **Reaction purity & inhibitors**
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B) PCR -----> Inputs , Rules , Outputs

Parameters (variables) :

1) Template DNA :

This is the material that would be used to perform PCR , without it you can't really do anything because you have no material .

- Typical Range → Any Range , Any sequence
- Why it matters → Defines target.

2) Primer length → Primers in general define the start point and end point of the region that would be amplified . So the length of the primer defines where you need it to bind.

- Typical ranges → 18 – 24 nucleotides mostly
Common and recommended usage range.

★ If primer length < 18 – 24 nucleotides (shorter than)
Specificity of binding to desired region is compromised.

★ If primer length > 18 – 24 nucleotides (longer than)
Slower annealing , Risk of secondary products , lower product yield.

- Why it matters → Primer lengths is one of the determinants of PCR success because it affects :

A) Specificity : primers must find only target sequence among billions or millions of bases in DNA.

B) If too short → Bind in many places by chance.

Rule for Specificity → Each additional base added , doubles the number of possible unique sequences.

B) Melting temperature (T_m) / Annealing behaviour :

it is the temperature at which half of the primer is bound to the template , depends heavily on length of the primer.

- Typical primer length regarding T_m → T_m falls in usable PCR ranges (50 – 65 °C).
- Too short primer length regarding T_m → They have low T_m , leading to weak or unstable binding.
- Too long primer length regarding T_m → Makes T_m high , thus it's harder to find proper annealing temperature.

C) Efficient hybridisation : This refers to the speed and efficiency of binding.

- **Short primers → Bind faster but with less specificity.**
- **Long primers → Take longer to bind and may hybridize with less efficiency at fixed annealing temperature.**

Efficiency rate → Ideal length = quicker and stronger binding to specific base pairing.

3) GC content in primers :

GC content refers to the percentage of guanine (G) and cytosine (C) bases in a DNA primer relative to its total base count. GC pairs form 3 hydrogen bonds , making them contribute more to duplex stability than A-T pairs that form 2 hydrogen bonds.

- Typical ranges → Should be about 40% to 60% GC. So for a 20-nucleotide primer , 8 to 12 of those bases should be G or C.
- Why GC content matters → GC content matters for many reasons.

A) Influences the binding strength of primer : Because GC have 3 hydrogen bonds. A higher GC content makes primer bind more stably to its template. Stability is directly tied to T_m as higher GC raises T_m , lower GC lowers T_m. So GC content is a major factor in setting PCR thermocycler conditions.

B) Ensures balanced annealing temperatures between primer pairs.

PCR uses two primers (forward and reverse). Their GC content and length determine their individual T_m values. For efficient amplification , primers should have similar T_m's (typically within a few °C of each other). If GC content

differ significantly , their T_m 's differ , they won't bind at same temperature , leading to failed or non-specific amplification.

C) High GC stretches (especially in G's) can form secondary structures like hairpins or pair with other primers (primer dimers), reducing primer availability and causing by-products.

Having a moderate GC content with balanced distribution helps minimize these problems.

D) Aids Specificity : Primers with too low GC may bind too weakly and non-specifically , whereas too much GC can make them bind too tightly to unintended sites in template. A balanced GC content helps ensure primers bind where they are supposed to.

★ If GC content is too low (< 40%) →

- Low duplex stability : primer doesn't bind strongly
- Low T_m : too weak binding can lead to no product or smeared amplification
- May require longer primers or altering sequence to raise T_m .

★ If GC content is too high (> 60% – 65%) →

- High T_m , very stable primer-template duplexes
- Can require higher annealing temperatures outside practical range , interfering with PCR efficiency.
- Secondary structure formation : GC-rich primers are more likely to make hairpins or primer dimers
- Complicated amplification of GC-rich regions : Entire PCR template with high GC can be hard to denature and amplify reliably , sometimes requiring additives or specialized enzymes.

4) Annealing Temperatures :

In PCR , the annealing step is when primers bind (anneal) to the single stranded DNA template after denaturation. The annealing temperature (T_a) is the temperature at which this binding occurs , it's crucial because it determines whether primers bind correctly and specifically to their target sequence or no.

- Typical range → Unlike denaturation and extension temperatures , the annealing temperature is variable and depends on the primers' melting temperature (T_m).

- Common practical range is 45 – 65 °C.
- Often , it is set 2 – 5 °C below the lowest primers' T_m in the primer pair.
- Exact values vary based on primer length , GC content , and design.

This flexibility ensures primers bind stably yet specifically to the template during reaction.

- Why it matters → Annealing temperature matters for many reasons.

A) Balances specificity and efficiency : Annealing temp determines how tightly primers bind.

- High T_a (closer to primer T_m) : Favors only stable and perfectly matched primer-template duplexes.

(High specificity)

- Low T_a : Allows primers to bind even with mismatches.
(Lower specificity and higher risk of non-target amplification)

★ That's because at higher temperatures unstable or mismatched matches melt , whereas correct matches persist.

B) Primer binding strength depends on sequence :

Annealing temperature is tied to the melting temperature (T_m) of primers , which is determined by :

- Primer length
- GC content (higher GC = higher T_m)
- Sequence composition

★ Setting T_a relative to T_m ensures primers anneal strongly enough for efficient extension , but not so weakly that they bind nonspecifically.

★ If annealing temperature is too low relative to T_m :

- primers can bind imperfectly
- leads to nonspecific amplification – extra or unexpected PCR bands
- primer dimers and other artefacts may form more easily
- yield of the desired product may be reduced because polymerase also amplifies incorrect product

This is because lower temperature allows partial or weak primer binding , not just the perfectly matched site.

★ If annealing temperature is too high :

- primers may not bind efficiently at all
- PCR yield drops or no product at all.
- Even if primers bind , it may be less stable and impede extension.

A too high T_a means even correct primer – template matches may not remain duplexed long enough for polymerase to extend.

★ Use T_m as your guide :

- A general rule of thumb from primer design is :
Set annealing temperature around 2 – 5 °C below the primers' T_m .
(especially the lowest T_m of the pair)

5) Mg^{2+} Concentration : Magnesium ions are one of the most critical chemical components of a PCR mix because they act as an essential cofactor for thermostable

DNA polymerase , like Taq polymerase . PCR reactions cannot proceed without Mg^{2+} because the enzyme needs it to function.

- Typical range → 1.5 – 2.5 mM of Mg^{2+} is common optimal range in standard PCR conditions.
- In many protocols the usable range is about 0.5 – 5.0 mM depending on polymerase , buffer , and template complexity.

Why Mg^{2+} Matters in PCR

Mg^{2+} has multiple essential biochemical roles in PCR:

◆ A. Cofactor for DNA Polymerase Activity

Mg^{2+} directly participates in the catalytic mechanism of DNA polymerases. It binds at the polymerase active site and coordinates with:

- the 3'-hydroxyl (3'-OH) of the primer and
- the incoming dNTPs

to facilitate phosphodiester bond formation. Without Mg^{2+} , the polymerase is largely inactive and amplification fails.

B. Stabilizes DNA–dNTP Complexes

dNTPs carry negative phosphate charges, which repel the DNA template. Mg^{2+} binds (chelates) to dNTPs and the DNA backbone, neutralizing negative charge and helping correct alignment of the substrate for polymerization.

◆ C. Influences Primer Annealing and DNA Stability

Mg^{2+} also interacts with the DNA duplex. Higher Mg^{2+} concentrations can increase the melting temperature (T_m) of double-stranded DNA and make primer–template binding more stable. In effect, Mg^{2+} makes it easier for primers to stick to the template, which changes how specific and efficient the PCR will be.

◆ D. Affects Fidelity and Specificity

Mg^{2+} concentration strongly influences:

- Yield (how much product you get),
- Specificity (whether only the target sequence is amplified),
- Fidelity (accuracy of nucleotide incorporation).

Too little can reduce polymerase function; too much can decrease stringency.

✗ A. Too Little Mg^{2+}

When magnesium concentration is too low:

- DNA polymerase activity *drops dramatically*, because the cofactor is insufficient.
- Primers may fail to bind effectively to the template.
- Result: low product yield or complete PCR failure.

This happens because Mg^{2+} is required both for enzyme catalysis and for stabilizing primer–template interactions.

✗ B. Too Much Mg^{2+}

When magnesium concentration is too high:

- Mg^{2+} can *over-stabilize primer–template interactions*, including mismatched or non-specific sites.
- This reduces stringency and leads to non-specific amplification — extra bands or smears on a gel.
- Primer dimers can form more readily because Mg^{2+} neutralizes negative charge and encourages random base pairing.
- High Mg^{2+} can also reduce polymerase fidelity, increasing misincorporated bases.

Why Mg^{2+} Must Be Optimized

Mg^{2+} does not act alone — its effective concentration depends on other components such as:

- dNTP concentration (dNTPs chelate Mg^{2+} and lower free Mg^{2+} available)
- Buffer composition
- Template complexity
- Presence of chelators like EDTA (which bind and remove Mg^{2+})

Because all these affect free Mg^{2+} levels, many labs perform a Mg^{2+} gradient experiment — testing a series of $MgCl_2$ concentrations to find the best balance of yield and specificity for a particular primer/template set.

6)PCR Cycles: A PCR cycle is one complete round of:

1. Denaturation
2. Annealing
3. Extension

Each cycle *theoretically doubles* the amount of target DNA present — this is the exponential amplification principle behind PCR. With each cycle the number of copies can roughly follow a 2^n pattern before reagents start limiting the reaction.

Typical Range of PCR Cycles

Most standard PCR protocols use a cycle range of:

✓ Common recommended range: ~25–35 cycles

- This range is effective for many research and analytic applications.

✓ Extended range: ~20–40 cycles

- Lower end (20–25): Enough when template concentration is high.
- Upper end (35–40): Sometimes needed when starting template is very low.

✓ Rare situations: >40 cycles

- Some specialized PCRs (like nested PCR or very low-copy templates) can go above 40, but this is *not standard* for routine amplification because of technical limitations.

Why Cycle Number Matters

◆ A. Controls Amplification Yield

Each PCR cycle ideally doubles the amount of target DNA.

- ~10 cycles = ~1,000-fold amplification
- ~20 cycles = ~1,000,000-fold
- ~30 cycles = ~1 billion-fold amplification

However, *in reality* efficiency isn't perfect, and reaction components become limiting after many cycles.

◆ B. Prevents Plateau Phase

After enough cycles:

- Species of DNA being amplified compete for primers and polymerase
- Reagents (primers, dNTPs, polymerase) get depleted
- Amplification enters a plateau phase, where additional cycles yield little or no new product.

So the cycle count also determines where the reaction falls on its amplification curve and how much usable product you get.

✗ Too Few Cycles

If you do too few cycles:

- Target DNA may not reach a detectable level – especially if initial template is low.
- Results can be weak bands or no product on a gel.
- May lead to false negatives in diagnostic or low-copy detection settings.

This happens because exponential amplification did not proceed far enough to accumulate product.

✗ Too Many Cycles

Increasing cycles beyond the optimal range leads to several problems:

◆ A. Non-specific Amplification

- Extra cycles increase the likelihood that primers will anneal nonspecifically and amplify unintended sequences.

◆ B. Primer-Dimers & Artefacts

- More cycles give primers more opportunities to bind to each other, forming primer-dimers and other artefacts that compete with your true product.

◆ C. Reagent Depletion & Plateau

- Reaction components become limiting after many cycles, reducing the efficiency of doubling.

◆ D. Lower Fidelity & Accumulated Errors

- Prolonged cycling increases the chance of polymerase errors accumulating in the product, which matters in sequencing or cloning applications.

7) PCR Buffers : In PCR, a buffer is the chemical solution that creates an *optimal environment* for the DNA polymerase enzyme to function throughout thermal cycling. It regulates pH, ionic strength, and provides supportive ions and stabilizers so the enzyme can synthesize DNA efficiently and specifically. Buffers typically contain:

- Tris-HCl — main buffering agent
- Monovalent salts, like KCl
- Divalent salts, mainly MgCl_2 (cofactor, often handled separately)
- Optional stabilizers or enhancers (e.g., BSA, detergents)

Typical Range

pH and Buffering Agents

- PCR buffers generally maintain pH ~8.0–9.5 at room temperature, stabilizing the chemical environment for the polymerase, which is *highly pH sensitive*.
- Tris-HCl is the core buffer, often supplied as a 10× stock that brings the final reaction buffer to ~10–50 mM Tris after dilution.

Monovalent Salts

- **Potassium chloride (KCl)** is commonly included to adjust ionic strength and help primers anneal by neutralizing negative charges on DNA. A common final concentration is around 25–100 mM KCl.

Divalent Ions

- Mg^{2+} is technically counted as part of buffer composition, though it's often optimized separately. Typical final MgCl_2 range is **1.0–5.0 mM**, with many standard protocols using ~1.5–2.0 mM.

Why Buffers Matter

A. Maintain Optimal pH

DNA polymerases have very narrow optimal pH ranges. For widely used Taq polymerase, activity is highest near pH ~8.0–8.8, and deviations reduce or disable enzyme function. Buffers resist pH changes caused by temperature cycling.

B. Ionic Environment for Primer Annealing

- Monovalent ions like K^+ reduce electrostatic repulsion between negatively charged DNA strands, helping primers bind specifically to the target.

C. Stabilize Enzyme & DNA

- Divalent cations like Mg^{2+} are essential *cofactors* for polymerase catalytic activity and help DNA duplex stability, though Mg optimization is usually discussed separately.

D. Improve Specificity & Yield

The right buffer composition helps ensure that only correct target amplification occurs — low nonspecific products, minimal primer dimers, and robust yield.

4. Types of Buffers & Additives

1) Standard PCR Buffer

- Core components: Tris-HCl + KCl + $MgCl_2$
- Common pH: 8.0–9.5
- Typical concentrations:
 - Tris: ~10–50 mM
 - KCl: ~25–100 mM
 - $MgCl_2$: ~1.5–2.0 mM (often optimized separately)

This is the most widely used buffer for standard genomic PCR.

2) High-Specificity Buffers

Some buffers include:

- Ammonium sulfate $(NH_4)_2SO_4$ — can enhance specificity by destabilizing mismatches
- Alternative salts — adjust ionic strength to favor specific primer annealing

✗ pH Out of Range

- If the buffer pH shifts too acidic or basic, DNA polymerase activity drops sharply, potentially **killing enzyme activity**. DNA also becomes less stable, leading to poor amplification or degradation of template/primers.

✗ Inappropriate Ionic Strength

- Too little monovalent salt (like KCl) → **poor primer annealing** and low yield.
- Too much salt → **reduced specificity** and possible mis-priming.

✗ Incorrect Mg^{2+} Levels

Although Mg^{2+} is often discussed separately, it's part of buffer performance:

- Too little Mg^{2+} → enzyme underperforms, little/no product.
- Too much Mg^{2+} → nonspecific products, reduced fidelity.

✗ Incompatible Additives

- Some additives (like excessive DMSO) can inhibit polymerase if overused — so additives must be *qualified for PCR*.

Incorrect buffer compositions are among the **most common causes of failed or low-quality PCRs** and are often checked first in troubleshooting.

8) Polymerase Enzyme in PCR : PCR uses a thermostable DNA-dependent DNA polymerase — an enzyme that can:

- survive repeated heating to ~95 °C
- synthesize new DNA strands from primers
- add nucleotides in the 5' → 3' direction

The classic enzyme is Taq polymerase (from *Thermus aquaticus*), but many engineered variants now exist.

Without polymerase → PCR cannot occur at all.

Typical Range

Polymerase is added in enzyme units (U), not molarity.

✓ Typical working range:

0.5 – 2.5 Units per 50 µL reaction

(1 U per reaction is the most common standard amount)

This range is recommended by most molecular biology manufacturers and PCR protocols.

Why Polymerase Amount Matters

Polymerase concentration controls:

Property	Why polymerase affects it
Yield	Determines how much DNA can be synthesized per cycle
Speed	Controls extension rate
Specificity	Excess enzyme amplifies mismatches
Fidelity	Some enzymes copy DNA more accurately
Product length	Determines max amplicon size

Types of PCR Polymerases

Polymerase Type	Property	Used for
Taq polymerase	Fast, no proofreading	Routine PCR
High-fidelity polymerases (Pfu, Q5, Phusion)	Have 3'→5' proofreading	Sequencing, cloning
Hot-start polymerases	Inactive until heated	Prevent primer dimers
Long-range polymerases	Amplify >10 kb DNA	Large genes/genomes
GC-rich polymerases	Work on GC-rich DNA	Difficult templates

✗ Too Little Polymerase

- Low or no PCR product
- Weak gel bands

- Reaction may fail completely

Because there isn't enough enzyme to extend all primed templates.

✗ Too Much Polymerase

- Non-specific amplification
- Primer-dimer formation
- Smearing on gel
- Reduced fidelity (more mutations)

Excess polymerase will amplify even weak or mismatched primer binding.

✗ Wrong Type of Polymerase

Problem	Result
Using Taq for sequencing	High mutation rate
Using non-hot-start enzyme	Primer dimers
Using standard enzyme on GC-rich DNA	No amplification
Using short-range enzyme for long genes	Truncated products

Polymerase Extension Speed (Important Range)

Polymerase	Extension speed
Taq	~1 kb / minute
Phusion/Q5	~15–30 sec per kb
Long-range polymerase	Up to 20 kb

This determines your extension time per cycle.

Polymerase controls whether DNA is copied correctly, efficiently, and specifically.

Too little → weak PCR.

Too much → messy PCR.

Wrong type → failed PCR.

9) dNTPs in PCR : dNTPs (deoxynucleotide triphosphates) are the building blocks that polymerase uses to *synthesize new DNA strands* during PCR.

They include:

- dATP
- dCTP
- dGTP
- dTTP

Without dNTPs, polymerase has no substrate to add — meaning **no DNA can be made**.

Typical Range of dNTP Concentration

PCR recipes usually specify dNTPs in micromolar (μM) concentrations.

✓ Typical final concentration per dNTP:

200 μM of each (dATP, dCTP, dGTP, dTTP) in the reaction.

✓ This means total dNTP concentration is ~800 μM (0.8 mM) when all four are present.

✓ Some protocols can use a *range* of 100–250 μM each depending on the template or enzyme used.

Why dNTP Concentration Matters

A) Substrate for DNA Polymerase

dNTPs are the *raw materials* the polymerase incorporates into the new DNA strand.

- Too little → polymerase *runs out of building blocks* and stops prematurely.
- Enough → efficient, complete DNA synthesis.

B) Affects Polymerase Function and Fidelity

Balance matters. Polymerase works best when the four nucleotides are present in roughly equal amounts.

- If one dNTP is limiting → misincorporation and stalling become more likely.

- Unequal pools can bias synthesis and increase errors.

C) Influences Buffer Chemistry and Magnesium Availability

dNTPs bind (chelate) magnesium ions (Mg^{2+}). Because Mg^{2+} is essential for polymerase activity:

- Higher dNTP levels reduce free Mg^{2+} in solution.
- That changes how magnesium must be optimized in your buffer.

So dNTP levels and Mg^{2+} levels are closely linked when optimizing PCR.

✗ Too Low dNTPs

- Polymerase stalls early
- Incomplete extension
- Low yield or no visible product
- Bands on gels may be weak or missing

This happens because polymerase simply uses up all of the substrates before cycles finish.

✗ Too High dNTPs

- Increased rate of misincorporations
- Lower fidelity (more mistakes)
- May increase non-specific products
- Requires more Mg^{2+} to compensate

High dNTP levels chelate Mg^{2+} , which lowers effective Mg^{2+} available for polymerase. When Mg^{2+} is too low due to chelation:

- ✓ The reaction becomes less specific
- ✓ Polymerase binding and extension suffer

This is a common reason high dNTP + low Mg^{2+} combinations fail.

Putting It All Together — Practical PCR Rules

Parameter	Typical Value	What It Affects
dNTP each	100–250 μM	Polymerase substrate
Total dNTP	~800 μM	Overall synthesis capacity
Mg ²⁺	1.5–2.5 mM	Polymerase cofactor

- ✓ Always add dNTPs *before* Mg²⁺ optimization — since they change free Mg²⁺ levels.
- ✓ Use equimolar dNTPs for best polymerase fidelity.

2) Gel Electrophoresis:

Introduction

DNA fragments cannot be seen directly, even after an experiment succeeds. Gel electrophoresis is a technique that **separates DNA fragments by size** and makes them visible.

When an electric field is applied, DNA moves through a gel matrix — **smaller fragments travel farther, larger ones move more slowly**.

The final pattern of bands reveals what DNA is present and how it differs between samples.

In this lab, you will load DNA samples, run the gel, and interpret the results to understand what your experiment actually produced.

A) Scope :

A) Scope

Why is gel electrophoresis used?

After many molecular biology experiments (especially PCR), you end up with **DNA fragments**, but you cannot see them directly.

Gel electrophoresis is the method used to:

- **Visualize DNA**
- **Estimate fragment size**
- **Compare samples**
- **Verify whether an experiment worked**

DNA fragments are separated based on **size** by pulling them through a gel matrix using an **electric field**.

Smaller fragments move faster.

Larger fragments move slower.

Gel electrophoresis does **not create DNA** – it only **reveals what already exists**.

What questions does gel electrophoresis answer?

Gel electrophoresis answers questions like:

- Did my PCR produce DNA?
- Is the fragment the expected size?
- Are there multiple products?
- Is there contamination?
- Are my samples comparable?

It turns invisible molecular outcomes into **interpretable visual evidence**.

Where students fail most (key educational value)

These failures are extremely common and **perfect for a virtual lab**:

1. **Misunderstanding band size**

- Thinking brighter = correct
- Ignoring fragment length

2. **Ignoring the DNA ladder**

- Not using it as a size reference
- Guessing band size by position alone

3. **Misinterpreting smears**

- Assuming smears mean “a lot of DNA”
- Not recognizing degradation or nonspecific amplification

4. **Confusing primer dimers with real products**

- Especially after PCR
- Small bands mistaken for success

5. **Assuming gels are qualitative only**

- Forgetting that intensity carries information
- Overtrusting visual results without controls

B) Gel ---> Inputs , rules , outputs :

1) Agarose concentration (%)

- **What:** Percent agarose in the gel — controls pore size of the matrix.
- **Typical / default:** **0.5% → 2.0%** depending on fragment sizes; common ranges: 0.5% (large fragments), 0.8–1.2% (broad use for 200 bp–10 kb), 1.5–2.0% (small fragments <500 bp). See NEB resolution table for mapping of % → size range.
- **Why it matters:** Lower % = larger pores → better separation of large fragments. Higher % = smaller pores → better resolution of small fragments. Wrong % causes poor separation or overlapping bands.
- **Troubleshooting / coding hints:** Expose to user as discrete slider or presets (e.g., “low, medium, high” or 0.5 / 0.8 / 1.0 / 1.5 / 2.0). Internally map: $\text{gel_pore_size} \propto 1/\text{\%agarose}$ and use this to affect migration resolution (band spread / separation). Flag “wrong gel for fragment size” when an expected band overlaps neighbors.

2) Running buffer (TAE vs TBE vs alternatives)

- **What:** Ionic solution that conducts current and maintains pH during runs.
- **Typical / default:** **1x TAE** (good for routine separation and downstream enzymatic recovery) or **1x TBE** (better buffering capacity for high-voltage/long runs and sharper resolution for small fragments). Some labs use sodium-borate (SB) for faster runs.
- **Why it matters:** Buffer ionic strength affects conductivity, migration speed (DNA runs ~10% slower in TBE vs TAE), heat generation, and how long the buffer remains effective. TBE has higher buffering capacity; TAE is more compatible with enzymatic downstream steps.
- **Troubleshooting / coding hints:** Give preset choices (TAE / TBE). Internally adjust `migration_rate` and `heat_risk` by buffer: e.g., TBE → slightly slower migration, better resolution for small fragments; TAE → faster migration, lower buffering capacity. If user runs long/time high voltage with TAE, show “buffer exhausted → smeared bands” warning.

3) Voltage (V) and Voltage per cm (field strength)

- **What:** Electrical potential applied across the gel; effective field strength is V divided by distance between electrodes (V/cm).
- **Typical / default:** **~4–10 V/cm** is a practical guideline for agarose gels (many protocols recommend 4–10 V/cm for routine DNA separations; lower for very large fragments). For a standard gel with electrodes 10 cm apart, that means 40–100 V.
- **Why it matters:** Higher voltage speeds runs but increases heat and can distort bands (smearing / smiling) or reduce resolution. Lower voltage gives better resolution but takes longer.
- **Troubleshooting / coding hints:** Provide a slider or presets (low/medium/high). Internally compute $V_{\text{per_cm}} = V / \text{electrode_distance}$. If $V_{\text{per_cm}} > 10$ and gel % high, trigger `overheat_distortion` outcome; if $V_{\text{per_cm}} < 3$ and target size small, show `poor_resolution`. Add visual feedback: band sharpness vs run time.

4) Run time / migration distance

- **What:** How long the current is applied or how far bands travel.
- **Typical / default:** Depends on voltage and fragment size — run until dye front reaches a target distance (e.g., 2/3 down gel). Tables map size → expected migration distance at standard conditions.
- **Why it matters:** Under-run → bands stay high (large apparent size); over-run → small fragments run off the gel. Run time interacts with voltage & gel % to determine band positions.
- **Troubleshooting / coding hints:** In simulation, compute migration using distance = $f(\text{fragment_size}, \text{gel_}\%, \text{V/cm}, \text{time})$. Cap max migration to gel length; if band reaches edge, mark as “run off”. Offer a real-time progress indicator (dye front position) to help users stop at the right time.

5) DNA ladder / size marker

- **What:** A mixture of DNA fragments of known sizes used as a ruler.
- **Typical / default:** Always include one lane with a ladder appropriate to expected fragment sizes (e.g., 100 bp ladder for small fragments; 1 kb ladder for larger fragments). Manufacturer-specific loading recommendations exist (amount per lane).
- **Why it matters:** Without a ladder, size estimation is unreliable. Ladder intensity and appropriate ladder choice determine the user’s ability to estimate fragment sizes accurately.
- **Troubleshooting / coding hints:** In UI require/encourage selecting a ladder before running. Internally use ladder band positions to calibrate bp ↔ distance mapping per run (since buffer/voltage/gel% change migration scaling). If user omits ladder, show “size estimation unavailable” and reduce confidence in size readout.

6) Sample amount (ng per well) & loading volume

- **What:** Quantity of DNA loaded into each well and the proportion of loading dye vs sample.
- **Typical / default:** Detectability and ideal loading vary with stain: with ethidium bromide detection ~10 ng per band is visible; many recommend **~10–100 ng per well** for PCR products; ladders have manufacturer-specified amounts. QIAGEN suggests up to 100 ng gives sharp bands with EtBr.
- **Why it matters:** Too little → no visible band. Too much → overloaded wells, smeared or merged bands. Proper loading dye/glycerol ensures sample sinks into well.
- **Troubleshooting / coding hints:** Model $\text{band_intensity} \propto \text{amount_loaded} \times \text{stain_sensitivity}$. Add checks: $\text{amount} < \text{detection_limit} \rightarrow \text{invisible}$, $\text{amount} > \text{overloading_threshold} \rightarrow \text{smear/merge}$. Expose a simple volume slider plus a preset “Auto-adjust to ladder” option.

7) Loading dye composition / tracking dyes

- **What:** Mix added to samples to add density and tracking dyes (bromophenol blue, xylene cyanol, orange G) and sometimes EDTA/glycerol.
- **Typical / default:** Standard loading dyes include glycerol (to sink) + tracking dyes. EDTA may be present to protect DNA.
- **Why it matters:** Tracking dyes let the user monitor progress (they migrate at approximate apparent sizes). Dyes also affect sample sinking and enzymatic compatibility.
- **Troubleshooting / coding hints:** Simulate dye front positions (use dye migration references) and show them so users can stop the run appropriately. If loading dye omitted, warn that sample may float and not enter well.

8) Staining & visualization (Ethidium Bromide vs SYBR Safe etc.)

- **What:** Chemical stains that intercalate or bind DNA and fluoresce under UV/blue light.
- **Typical / default:** Historically **ethidium bromide (EtBr)**; safer alternatives like **SYBR Safe** (Invitrogen/Thermo) are widely used. Sensitivity varies; SYBR Safe is comparable or better in many contexts and is less mutagenic.
- **Why it matters:** Stain determines detection sensitivity and safety; imaging method (UV vs blue light) and exposure affect band intensity readout. Some stains are added to gel before casting or used as post-stain.
- **Troubleshooting / coding hints:** Let user choose stain (EtBr / SYBR / post-stain). Internally map stain_sensitivity and safety_flag (e.g., EtBr: high sensitivity, mutagenic; SYBR: high sensitivity, safer). Adjust detection limits accordingly.

9) Gel thickness & well size (well capacity)

- **What:** Physical dimensions of the gel (thickness, well width/depth) affecting volume capacity and heat dissipation.
- **Typical / default:** Standard gels ~3–5 mm thick; wells sized to hold typical loading volumes (e.g., 10–50 μL).
- **Why it matters:** Thicker gels hold more sample but may heat slower; very thin gels are fragile. Well size limits maximum sample loaded before overflow and affects band shape.
- **Troubleshooting / coding hints:** Provide a preset well capacity (e.g., 10 μL , 20 μL , 50 μL). If user loads more than capacity, simulate overflow and lane distortion.

10) Temperature / overheating & buffer heating

- **What:** Heat generated during runs due to current; depends on voltage, buffer ionic strength, and gel geometry.
- **Typical / default:** Avoid excessive heating; higher volts increase heat risk. Cooling or lower V/cm can reduce distortion.
- **Why it matters:** Overheating causes smiling bands, diffusion, and can melt gel or cause band distortion. Some protocols recommend running in a cold room or using buffer recirculation for long/high-voltage runs.
- **Troubleshooting / coding hints:** Internally compute heat = $f(V, \text{buffer_conductivity}, \text{run_time})$ and if above threshold set smile_distortion = true and reduce band sharpness.

Provide a “cooling” option (lower V or ice block) to recover resolution.

11) Causes of smears / degraded DNA / artifacts

- **What / Why:** Smears can result from degraded template, overloading, degraded samples, salt contamination, incomplete digestion, or running conditions. Primer-dimers from upstream PCR appear as small low-molecular-weight bands.
- **Troubleshooting / coding hints:** Build a mapping from upstream PCR errors to gel patterns: e.g., primer_dimer → small $\sim <100$ bp band; degraded_template → smear across lane; overload → smear/high intensity + distorted bands. Use this mapping to generate explanatory feedback messages.

12) Quantification limits & semi-quantitative interpretation

- **What:** Gel band intensity is only semi-quantitative; intensity depends on amount, staining efficiency, and exposure.
- **Typical / default:** Use ladders and standards for approximate quantification; accurate quantification needs image analysis and standards.
- **Why it matters:** Students often misinterpret brightness as absolute quantity. Gel readouts should be presented with confidence levels (approximate / qualitative).
- **Troubleshooting / coding hints:** Provide a “quantify” readout with caveats: $\text{estimated_ng} = \text{calibrated_from_ladder} \times \text{intensity_adjusted_by_stain}$. If user wants precise quantification, show a message recommending fluorometric/photometric methods.

Implementation notes (MVP vs Advanced)

- **MVP (what to expose to users):** Agarose % presets, buffer type (TAE/TBE), ladder selection, sample amount (low/medium/high), voltage presets (low/med/high with recommended V/cm shown), run time slider or “run until dye at X”, stain selection (EtBr/SYBR Safe), basic interpretation text + flagged issues (no ladder / overloaded / smear / primer-dimer). (References: NEB, Thermo Fisher, Addgene.)
- **Advanced (Phase 2):** Exact V/cm numeric input, Mg^{2+} /ionic strength effects (when simulating RNA/protein gels or enzymatic steps), buffer exhaustion modeling, quantitative band intensity calibration, real-time migration animation, temperature / heat-map simulation, and post-run analysis tools.

Quick checklist for each simulated run (for the engine)

1. Confirm ladder chosen → calibrate size-distance function.
2. Use agarose % → set pore_size modifier.
3. Compute field (V/cm) → derive base migration rate.
4. For each fragment, compute distance = $k * f(\text{size}, \text{pore_size}, \text{V/cm}, \text{buffer})$; clamp to gel length.
5. Compute intensity from amount_loaded $\times$ stain_sensitivity $\times$ transfer_efficiency. Flag >overload.
6. Apply heat/voltage distortion if heat > threshold → broaden bands / smile.

7. Output: band positions, sizes (estimate), intensity, and human-readable interpretation (success/possible primer dimers/smear/ladder mismatch).

3) Enzyme activity :

Introduction

Enzymes are biological catalysts that speed up chemical reactions essential for life.

Without enzymes, most reactions would occur too slowly to sustain cells.

Enzyme activity depends on conditions such as substrate availability, temperature, and pH. Small changes in these conditions can increase, reduce, or completely stop the reaction.

In this lab, you will control the reaction environment and observe how enzyme activity changes over time.

A) Scope

Why is enzyme activity studied?

Enzymes are biological catalysts — they speed up chemical reactions that would otherwise occur too slowly to sustain life.

Almost every cellular process depends on enzymes:

- Metabolism
- DNA replication
- Transcription & translation
- Signal transduction
- Digestion
- Detoxification

Studying enzyme activity allows us to:

- Understand how fast reactions occur
- Determine optimal conditions
- Detect inhibition or malfunction
- Compare enzyme efficiency

Enzyme experiments do not ask *“Is something present?”*

They ask *“How well does it work?”*

What questions does enzyme activity answer?

Enzyme activity experiments answer questions like:

- Is the enzyme functional?
- How fast is the reaction?
- What conditions maximize activity?
- What inhibits the enzyme?
- How does substrate concentration affect rate?

These experiments transform invisible molecular events into measurable reaction rates.

Where students fail most (key educational value)

Enzyme activity is conceptually simple but frequently misunderstood:

1. Confusing enzyme amount with enzyme efficiency
 - More enzyme $\neq$ better enzyme
 - Activity depends on conditions
2. Misunderstanding saturation
 - Thinking rate increases forever with substrate
 - Ignoring V_{max}
3. Ignoring environmental conditions
 - pH and temperature effects underestimated
4. Misinterpreting inhibition
 - Assuming inhibitors “destroy” enzymes
 - Confusing competitive vs non-competitive inhibition
5. Treating activity as binary
 - Thinking enzymes either “work” or “don’t”
 - Missing gradual rate changes

B) Enzyme Activity → Inputs, Rules, Outputs

1) Enzyme concentration

What it is → the amount (mass or molar) of active enzyme present in the reaction. Often expressed as $\mu\text{g/mL}$ or nM active sites.

Typical / default → depends on assay and enzyme; for basic educational assays use a range that shows clear rate change, e.g. low/med/high corresponding to $\sim 0.01\text{--}1\times$ the working amount used in lab protocols. In literature, enzyme concentrations are chosen such that initial rates are measurable but substrate is not immediately depleted.

Why it matters → initial reaction velocity (v_0) is directly proportional to enzyme concentration when substrate is in excess; increasing enzyme increases V_{max} (more active sites). At very low $[E]$ the reaction can become limited by enzyme availability and initial-rate assumptions break down.

Troubleshooting / coding hint → model discrete states: absent / low / optimal / high / saturating. If substrate $\gg K_m$ then rate $\propto [E]$; if substrate $\ll K_m$ then enzyme changes have smaller effect. In UI expose a simple slider or presets; internally compute $V_{max} \propto [E]$. Flag “no product” when enzyme = absent.

2) Substrate concentration

What it is → concentration of the enzyme’s reactant substrate(s) (typically μM – mM in assays).

Typical / default → choose values around the enzyme’s K_m : low ($\sim 0.1 \times K_m$), mid ($\sim 1 \times K_m$), high ($\sim 10 \times K_m$). For teaching, present Low/Medium/High tied to K_m so users see first-order → zero-order transition.

Why it matters → reaction rate rises with substrate concentration following Michaelis–Menten behaviour: initially approximately linear (first order), then plateaus toward V_{max} as the enzyme becomes saturated. K_m is the $[S]$ at half- V_{max} and gives substrate affinity information.

Troubleshooting / coding hint → implement Michaelis–Menten formula for initial rate: $v = (V_{max} * [S]) / (K_m + [S])$. Let users pick substrate relative to a displayed K_m value for immediate intuition. If $[S]$ very low → show “substrate-limited”; if $[S]$ very high and rate unchanged → show “saturation (V_{max}) reached”. Cite and use initial-rate assumption for short time windows.

3) Temperature

What it is → reaction temperature ($^{\circ}\text{C}$) controlling molecular kinetic energy and enzyme stability.

Typical / default → enzyme-specific. For mesophilic enzymes (e.g., human enzymes) optimal $\approx 37^{\circ}\text{C}$; bacterial enzymes vary (e.g., Taq polymerase optimal higher ~ 70 – 75°C). For teaching, provide presets: 4°C , 20°C , 37°C , 50°C , 70°C .

Why it matters → increasing temperature generally increases rate (Arrhenius effect) up to an optimum; above that thermal unfolding / denaturation reduces activity – denaturation is both temperature- and time-dependent. Observed “temperature optimum” is a balance between catalytic acceleration and loss of active enzyme over the assay interval.

Troubleshooting / coding hint → model rate as $\text{rate} = f_T(\text{Arrhenius}) \times \text{stability_factor}(\text{time}, T)$ where stability_factor decays faster at higher T (time-dependent denaturation). For MVP, a simple triangular curve (rises to T_{opt} then drops sharply) is enough; advanced mode can add time dependence so high- T short runs still show activity but long runs cause denaturation. Flag “denaturation risk” when T exceeds enzyme’s stability limit.

4) pH

What it is → acidity/alkalinity of the reaction medium (pH unit).

Typical / default → enzyme-specific optimum; many cytosolic enzymes $\approx \text{pH } 7$ – 8 , digestive enzymes different (pepsin $\approx \text{pH } 2$). For teaching provide presets: acidic (pH 3–4), neutral (pH

7), basic (pH 9).

Why it matters → pH affects ionization states of catalytic residues and substrate, folding and active-site geometry; activity typically peaks in a narrow pH range and falls off outside it — extreme pH can irreversibly denature protein structure.

Troubleshooting / coding hint → implement a bell-shaped activity vs pH function centered on enzyme_pH_opt; if pH deviates by >1–2 units reduce activity significantly; extreme values trigger denaturation flags. Allow “buffer choice” as an advanced control (phosphate, Tris, acetate) if you add buffer chemistry later.

5) Time (assay duration & sampling)

What it is → how long the reaction is run and when measurements are taken.

Typical / default → measure initial rate over a short window (where <5–10% substrate consumed) for Michaelis–Menten initial-rate analysis; full progress curves can be minutes to hours depending on enzyme. For teaching, let users run short (initial rate) and long (progress) modes.

Why it matters → early (initial) rates reflect enzyme kinetics (v_0) without complications (product inhibition, substrate depletion, enzyme instability). Long runs show product accumulation, substrate exhaustion, and time-dependent effects (denaturation, feedback).

Troubleshooting / coding hint → default to initial-rate reporting for quantitative outputs; provide a progress graph (product vs time) and an option to “extend run” to observe substrate depletion or denaturation. If users request K_m/V_{max} estimation, run multiple initial-rate simulations at different [S] and fit to MM equation (Phase 2 feature).

6) Inhibitors (type & concentration)

What it is → molecules that reduce enzyme activity; classes include competitive, non-competitive, uncompetitive, and mixed.

Typical / default → simple on/off inhibitor toggle for MVP; advanced: adjustable inhibitor concentration and type. Competitive inhibitors raise apparent K_m but not V_{max} ; noncompetitive reduces V_{max} ; uncompetitive reduces both in characteristic ways.

Why it matters → inhibition is central to pharmacology and regulation. Different inhibitor types produce distinct kinetic signatures that are teachable and diagnosable from rate vs [S] plots.

Troubleshooting / coding hint → implement three basic behaviors for MVP:

- competitive: $v = (V_{max} * [S]) / (K_m * (1 + [I]/K_i) + [S])$ — shows apparent K_m increase.
- noncompetitive: reduces V_{max} (e.g., $V_{max_eff} = V_{max} / (1 + [I]/K_i)$).
Show short text explanations and visual overlays comparing inhibited vs control runs.
Use Ramsay/NCBI reviews as background for mechanistic notes.

7) Cofactors / ionic strength / buffer composition (optional advanced parameter)

What it is → required metal ions (Mg^{2+} , Zn^{2+}), cofactors (NAD^+/NADH), and overall ionic environment.

Typical / default → either "cofactor present" or "absent" in MVP; advanced: numerical concentrations (mM). Many enzymes require specific metal ions at mM or μM levels.

Why it matters → cofactors can be absolutely required for activity (apo → holo transitions). Ionic strength and salts influence folding, substrate binding, and rate. Missing cofactors = no activity.

Troubleshooting / coding hint → include "cofactor toggle" that when off sets activity to zero for cofactor-dependent enzymes. For later depth, simulate competitive chelation (e.g., EDTA removes Mg^{2+}) and show corresponding activity drop.

Engine / implementation checklist (summary for CS friend)

1. Use Michaelis–Menten baseline for initial-rate behavior and V_{max} scaling with $[\text{E}]$.
2. Model temperature as rate multiplier $\times$ time-dependent stability decay (simple bell curve for MVP; add time-dependent denaturation later).
3. Model pH with a bell curve around enzyme_pH_opt; deviation reduces k_{cat} and can denature at extremes.
4. Implement inhibitor formulas for competitive / noncompetitive in MVP and provide comparative plots.
5. UI hooks: presets for common enzyme classes (human/catalase/Taq), sliders for Enzyme, Substrate, Temp, pH, Time, Inhibitor. Show product vs time and initial-rate fit.

Rules (internal logic)

- Enzymes lower activation energy but are not consumed
- Reaction rate depends on:
 - Enzyme concentration
 - Substrate availability
 - Environmental conditions
- Saturation exists — enzymes have a maximum reaction rate
- Denaturation permanently reduces or eliminates activity
- Inhibition reduces activity without changing substrate identity

Enzyme reactions are continuous and dynamic, not binary.

Outputs (what the user sees)

Primary Outputs

- Reaction rate (slow / optimal / fast)
- Amount of product formed
- Activity over time
- Visual indication of saturation or denaturation

Interpretable Results

- Increasing substrate without rate increase → enzyme saturation
- Sharp activity loss at high temperature → denaturation
- Reduced rate with inhibitor present → inhibition effect
- Flat reaction curve → missing enzyme or unsuitable conditions